

Miyoo Mini, gaming handheld with ssd202d #41

 on Nov 9, 2021 · 16 comments · 48 replies

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Miyoo Mini, gaming handheld with ssd202d #41

asoderq started this conversation in [General](#)

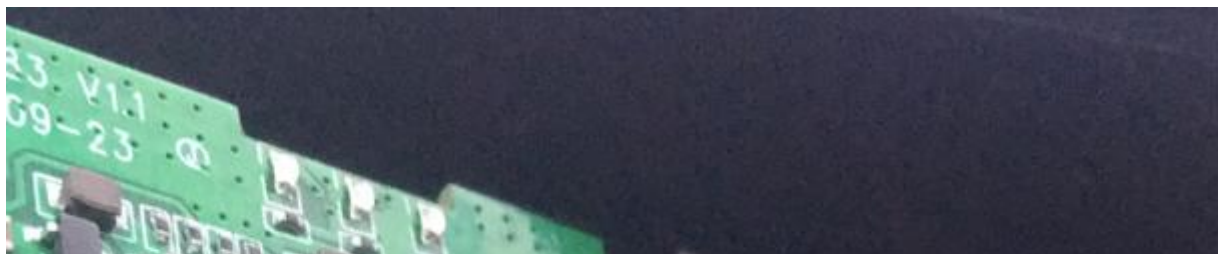


asoderq on Nov 9, 2021

...

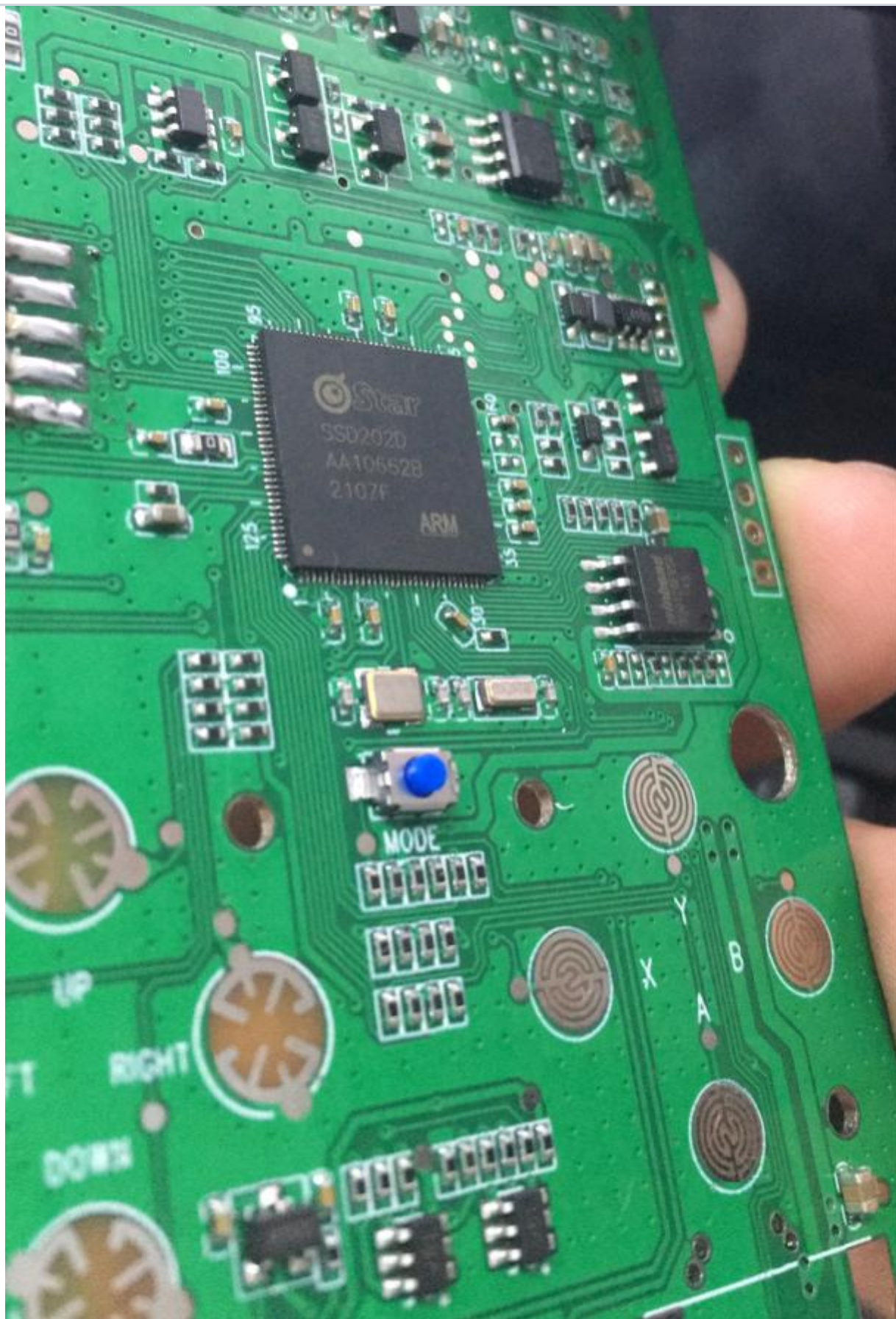
Hi,

I just wanted to make you aware that there is a tiny gaming handheld in the makings that uses the ssd202d. It should be released relatively soon. The only other thing I know about it is that it is supposed to have a 640x480 display.



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↑ 4



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fifteenhex on Nov 9, 2021

...

Thanks for the heads up. Hopefully I can get one of these and get the display etc working.



2 replies



asoderq on Nov 9, 2021

 ...

I will post a link to aliexpress when it is out. I am following it quite closely.



fifteenhex on Nov 9, 2021

 ...

Thanks! I'll keep an eye on taobao too. I guess I need to try harder at getting sound working in my kernel now. :)



Write a reply



fifteenhex on Nov 10, 2021

...

Found this after googling a bit:

<https://ezbuy.sg/product/51000217825092.html>

If that price is right and USD then the price is pretty good.



2 replies



fifteenhex on Nov 10, 2021

 ...

scratch that. Looks like it's a different thing :(



asoderq on Nov 10, 2021

 ...

That is the predecessor, called Bittboy. It uses an Allwinner F1Cxxx series. I am not sure which one.



Write a reply



asoderq on Nov 10, 2021

...

I realized that the SSD202D doesn't have a USB 'device' port, which is inconvenient for app development. Other similar devices I played around with uses this for Ethernet over USB so that you can remotely login

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know if it is possible in the USB standard to perform both charging function and USB 'host' on single connector?



12 replies



Show 7 previous replies



fifteenhex on Jan 4



USB-C on miyoo mini is only for charging

Doesn't seem so.. I traced it out and the USB-C connector looks like it's connected to the usb host on the SoC ->

https://twitter.com/linux_chenxing/status/1478270167134404609



fanoush 4 days ago



USB-C on miyoo mini is only for charging

Doesn't seem so.. I traced it out and the USB-C connector looks like it's connected to the usb host on the SoC -> https://twitter.com/linux_chenxing/status/1478270167134404609

hi @fifteenhex any news about getting usb host working over usb-c on miyoo mini?



fifteenhex 4 days ago



I don't think the USB port is able to supply power. You'd need to put together a mod to supply 5v on the usb-c port when a device is attached.



fanoush 4 days ago



I don't think the USB port is able to supply power. You'd need to put together a mod to supply 5v on the usb-c port when a device is attached.

Yes, that wouldn't be a problem if it otherwise worked. There are cables for phones like <https://www.aliexpress.com/item/33033635055.html> which can both charge the phone and attach some usb device to it.

I do have such cable but haven't actually tried as I expected this won't work out of box with linux kernel build currently in the device.



fanoush 4 days ago



I actually have few different similar cables. They work fine with raspberry pi 4 and Pi400 which also can do usb host on its usb-c power connector without providing power.



Write a reply

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A relatively new video about Miyoo Mini:

<https://www.bilibili.com/video/BV1Wh411b7L6>

Apparently they have put a PSP emulator on it, I didn't even know that there existed a software renderer PPSSPP.



2 replies



asoderq on Nov 17, 2021



No idea why they mention a bunch of other SoCs in the video but not the SSD202D.



asoderq on Nov 18, 2021



This video might just be speculations from someone. Best to take it with a grain of salt.



Write a reply



asoderq on Dec 2, 2021



It has been released now, and is available on aliexpress. Although I don't have taobao account, I heard it was sold out there. Here is a link to ali:

<https://www.aliexpress.com/item/1005003611677275.html>

but as usually you may find it elsewhere.



5 replies



fifteenhex on Dec 2, 2021



Thanks, Ordering some now..

Couldn't find them on taobao but I guess taobao isn't a good way to get them anyway as the agents usually refuse to ship stuff with batteries.



asoderq on Dec 2, 2021



I just noticed, be aware that this seller doesn't have any reviews...



fifteenhex on Dec 2, 2021



Getting money back from sellers that don't deliver is basically the only thing paypal is useful for these days :D.



asoderq on Dec 2, 2021



When you receive it please post high res PCB pictures :)

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sure. I'll probably take back and from scans to overlay and trace out, write a wiki page with where things are wired up and then write a devicetree.



Write a reply



asoderq on Dec 11, 2021



A video about the miyoo mini:

https://www.youtube.com/watch?v=IlyXE1-OPDI&ab_channel=AlwaysBeFun



5 replies



asoderq on Dec 13, 2021



Follow up video, apparently he bricked his miyoo mini with an update:

https://www.youtube.com/watch?v=SNcezdYFg1Y&t=1s&ab_channel=AlwaysBeFun



fifteenhex on Dec 13, 2021



He's getting a splash screen so that suggests it's getting to u-boot at least. I suspect it's possible to unbrick it without replacing the chip like he mentions..

First thing I'll do is dump out the original firmware just in case.



asoderq on Dec 13, 2021



I assume the internal memory is holding the bootloader. How would you go about to flash that? If the contents somehow gets corrupted, failed flashing etc, is it then bricked forever?



TriForceX on Dec 14, 2021



He's getting a splash screen so that suggests it's getting to u-boot at least. I suspect it's possible to unbrick it without replacing the chip like he mentions..

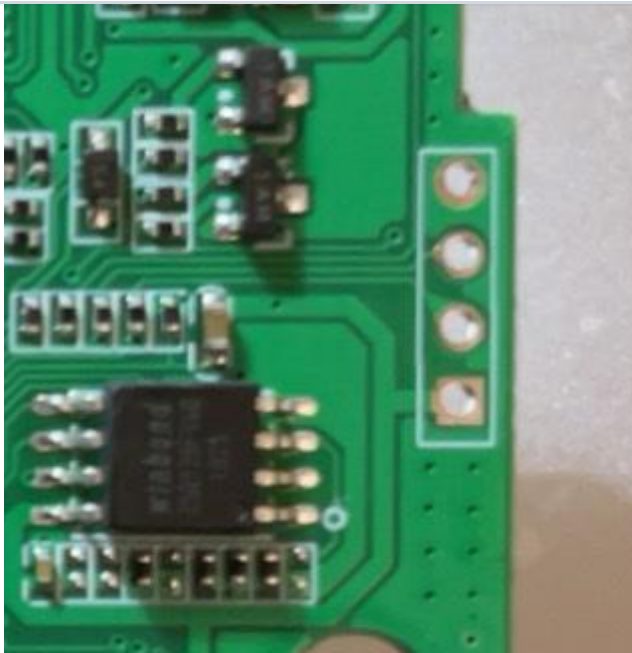
First thing I'll do is dump out the original firmware just in case.

i dont think a chip replacement is required, maybe just rewritting a backup into it (through uart) should be enough.

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fifteenhex on Dec 15, 2021



Yes, that should be PM_UART. If you have an i2c dongle you can write the flash directly through it. It's how I'll be backing mine up with they come



Write a reply



fifteenhex on Dec 12, 2021



Now on taobao: <https://item.taobao.com/item.htm?spm=a230r.1.14.16.562176f31emiWI&id=662148753424&ns=1&abbucket=5#detail>



0 replies



Write a reply



fifteenhex on Dec 15, 2021



I've started a wiki page for this here: <http://linux-chenxing.org/infinity2/miyoomini/>



3 replies



TriForceX on Dec 15, 2021



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on Nov 9, 2021 · 16 comments · 48 replies

haven't uploaded yet)

My plan is to add support to the console, similar to the work i did for first Miyoo consoles.

Im also preparing a package to update the SD contents (like proper configs for RetroArch) and the required build tools to ports APPS to Miyoo mini that the manufacturer sent to me.



fifteenhex on Dec 16, 2021



I guess they give you the old 4.x kernel to work with and the SDK binaries.

If we could get the sound working I think we could have everything you need working in our 5.16 kernel...

alas getting sound working is proving to be a massive pain.



TriForceX on Dec 18, 2021



I guess they give you the old 4.x kernel to work with and the SDK binaries. If we could get the sound working I think we could have everything you need working in our 5.16 kernel... alas getting sound working is proving to be a massive pain.

They refused to share any kernel / firmware source. I was able to get the SDK / toolchain only.



Write a reply



asoderq on Dec 16, 2021



Another review: https://www.youtube.com/watch?v=d7GXZ2wpA9U&ab_channel=TakiUdon



1



1

4 replies



fifteenhex on Dec 16, 2021



Playstation games seem to work better than I expected. I haven't used a ps1 emulator in ages but all of the comments about the miyoo mini I read said it's probably only good for 2D ps1 games. Seems to be running some 3D games at a playable level.



asoderq on Dec 16, 2021



Jupp, the dynamic re-compiler helps a lot. Without it would run like crap. I think the whole library would run full speed if it clocks to 1.5GHz.

EDIT: Also, the GTE (Geometry Transformation Engine) is written with NEON instructions.



fifteenhex on Dec 17, 2021



I've had the SSD202D boosting to 1.8GHz (assuming we are measuring it correctly..). Not sure if it would run like that constantly.

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asoderq on Dec 17, 2021



I am not that intimately familiar with PCSX-REARMED, but I don't think it benefits that much from multiple cores. PCSX4ALL, shares some heritage with PCSX-REARMED, is running fairly well on the single 1GHz Xburst core in the jz4770 and I believe it runs all games full speed on a single 1.5GHz Xburst core in the x1830. That is what I base my rationale on. Xburst should have slightly better DMIPS/MHz than A7 core, but I think PCSX REARMED with the NEON GTE is more optimized. Just my personal rough approximation.



Write a reply



asoderq on Dec 17, 2021



The display is apparently mounted 180 degree rotated. So the necessary 2d Graphics functions are rotation and hardware scaling.



1



0 replies



Write a reply



ddB0515 on Dec 18, 2021



got mine today from Ali so if anyone is interested I can dump SPI chip tomorrow and get more info (and maybe get boot log)



1



2 replies



TriForceX on Dec 18, 2021



If you can send me the dump the SPI i would appreciate, im going to support Miyoo Mini too



fifteenhex on Dec 18, 2021



Getting some big high resolution photos of the back and front of the PCB would be really good. We can start tracing out the pins from the SoC. Output from booting and stuff like the output of /sys/kernel/debug/gpio would be good too.



1



Write a reply

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on Nov 9, 2021 · 16 comments · 48 replies

Here is attached boot log

[boot.log](#)

Dumped SPI

```
flashrom v1.2 on Linux 5.11.0-43-generic (x86_64)
flashrom is free software, get the source code at https://flashrom.org
```

```
Using clock_gettime for delay loops (clk_id: 1, resolution: 1ns).
Found Winbond flash chip "W25Q128.V" (16384 kB, SPI) on ch341a_spi.
Reading flash... done.
```

[MiYoo283v1.1.zip](#)

And log from /sys/kernel/debug/gpio

```
/ # cat /sys/kernel/debug/gpio
gpiochip0: GPIOs 0-90, gpio:
gpio-1  (          |GPIO Key Up          ) in hi
gpio-5  (          |GPIO Key Right       ) in hi
gpio-6  (          |GPIO Key B          ) in hi
gpio-7  (          |GPIO Key Y          ) in hi
gpio-8  (          |GPIO Key A          ) in hi
gpio-9  (          |GPIO Key X          ) in hi
gpio-10 (          |GPIO Key START      ) in hi
gpio-11 (          |GPIO Key SELECT     ) in hi
gpio-12 (          |GPIO Key MENU       ) in hi
gpio-13 (          |GPIO Key L          ) in hi
gpio-14 (          |GPIO Key L2         ) in hi
gpio-47 (          |GPIO Key R2         ) in hi
gpio-59 (          |sysfs              ) in hi
gpio-69 (          |GPIO Key Down       ) in hi
gpio-70 (          |GPIO Key Left       ) in hi
gpio-72 (          |                   ) out lo
gpio-86 (          |GPIO Key POWER      ) in lo
gpio-90 (          |GPIO Key R          ) in hi
/ #
```

And Front and back images (as best I could take them)

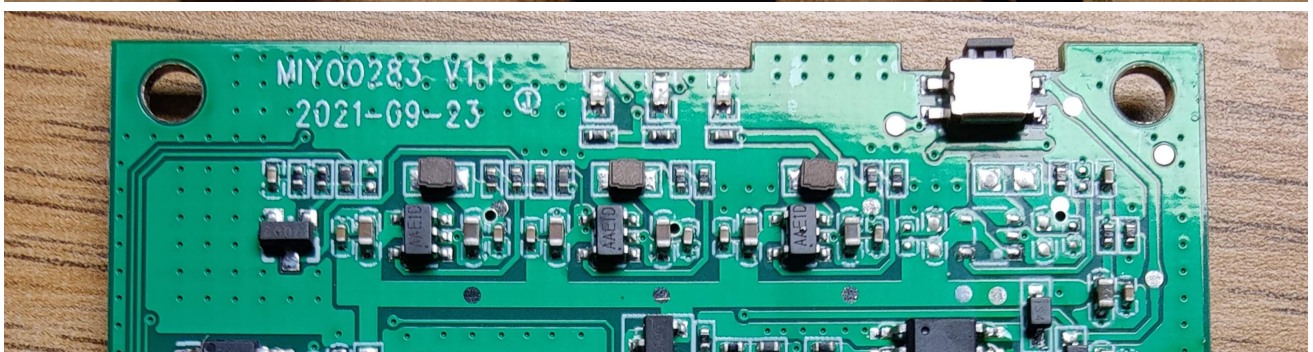
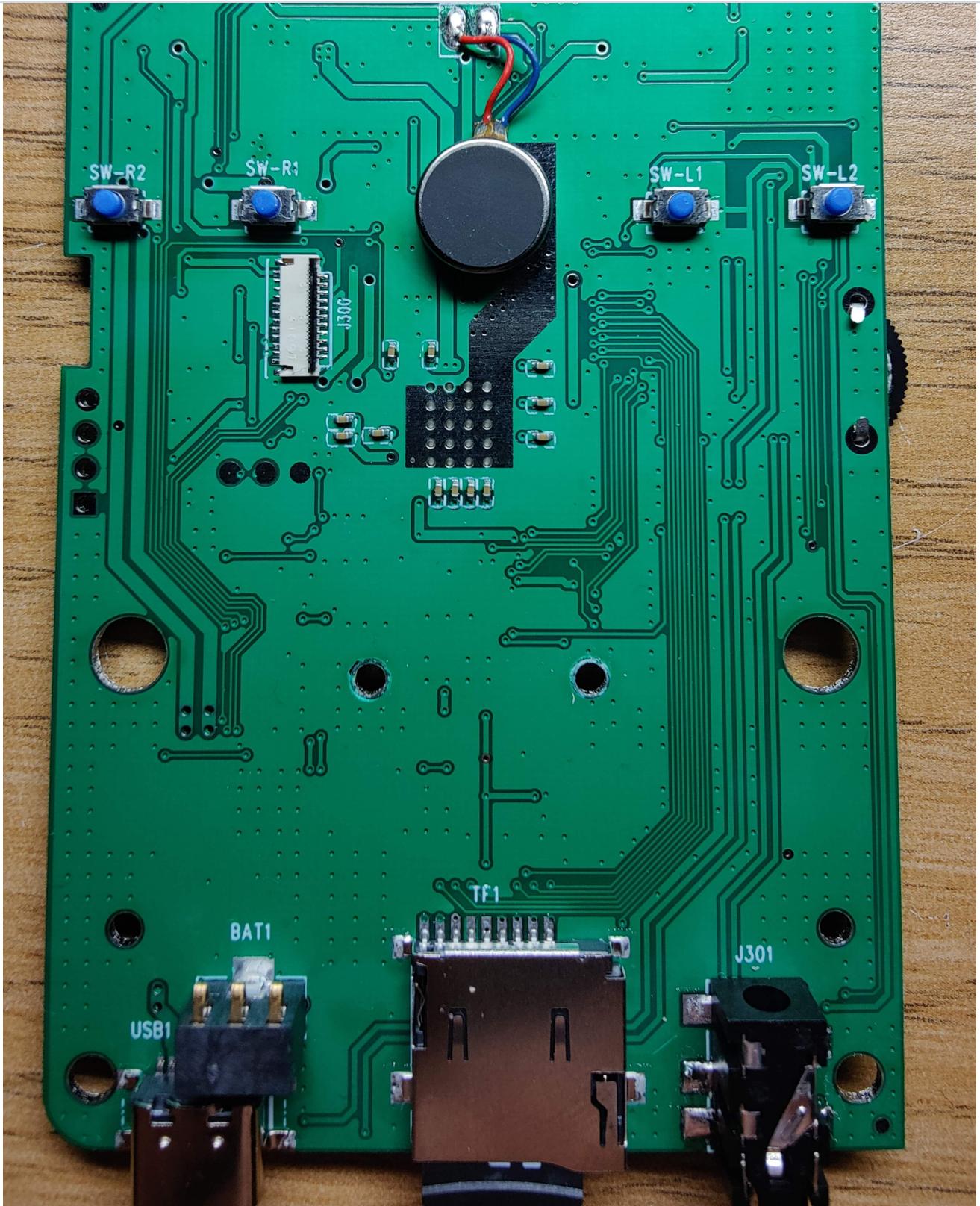
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on Nov 9, 2021 · 16 comments · 48 replies

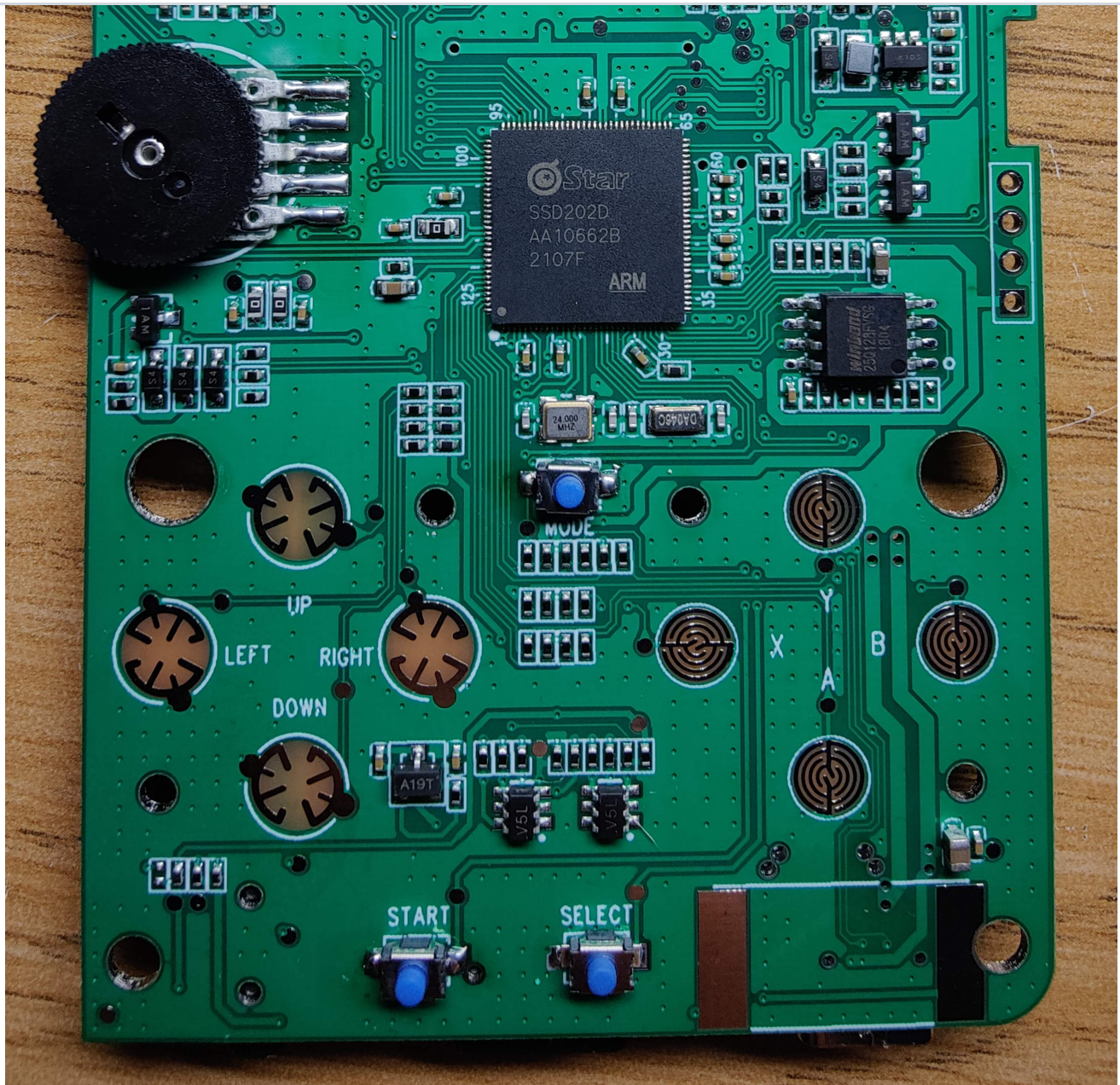
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 on Nov 9, 2021 · 16 comments · 48 replies



5 replies



fifteenhex on Dec 19, 2021



Do you mind if I stick these photos on the wiki?



TriForceX on Dec 19, 2021



thanks @ddB0515 ive uploaded your stuff in the miyoo mini section too
<https://github.com/TriForceX/MiyooCFW/wiki/Miyoo-Mini>



ddB0515 on Dec 19, 2021



Do you mind if I stick these photos on the wiki?

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what's baud rate of UART? I have tried 9600, 115200...



ddB0515 on Jan 3



I have used 115200 and worked for me, check RX-TX connections that was issue for me usually



Write a reply



fifteenhex on Jan 4



Finally got mine today.. working on getting mainline running now :)



0 replies



Write a reply



fifteenhex on Jan 4



u-boot env

```
SigmaStar # printenv
SdUpgradeImage=miyoo283_fw.img
baudrate=115200
bootargs=console=ttyS0,115200 root=/dev/mtdblock4 rootfstype=squashfs ro init=/linuxrc LX_MEM=0x7
bootcmd=gpio output 85 1; bootlogo 0 0 0 0 0; gpio output 4 1; mw 1f001cc0 11; gpio out 8 0; sf p
bootdelay=0
cpu_part_start=14270000
dispout=K101_IM2BVL
ethact=sstar_emac
ethaddr=00:30:1b:ba:02:db
filesize=be44
miyoo_upgrade=gpio output 4 1; bootlogo 1 0 0 0 0;sdstar;
miyoo_version=202111201656
sdfw_version=202110071805
sf_kernel_size=200000
sf_kernel_start=60000
sf_part_size=20000
sf_part_start=270000
stderr=serial
stdin=serial
stdout=serial
usb_folder=images
```

Environment size: 913/4092 bytes

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0 replies



Write a reply



asoderq on Jan 4



Yet another review: https://www.youtube.com/watch?v=Kx5W1BNGVU4&ab_channel=RetroGameCorps
Main complaint seems to be a "hazy" screen, I wonder if that could be fixed...



5 replies



whmzsu 28 days ago



The latest version had improved the screen similar to OGS tech.



fifteenhex 28 days ago



Looks like the battery changed too? The photos on aliexpress now have a battery with a cable.



erazemk 28 days ago



Looks like the battery changed too?

It has, it's now a 2000mAh one iirc.



fifteenhex 25 days ago



I bought two of the new ones. Will add some photos of the guts when they arrive.



fifteenhex 17 days ago



Screen seems to be the same.. but the new ones I bought do seem to be clearer than the original ones I got.

PCB differences are minimal. Just a connector for the new battery.



Write a reply



fifteenhex on Jan 6



Making good progress:

All of the buttons are mapped out but left and down don't register for some reason. Probably some pinctrl stuff.

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↑ 1



👍 4

1 reply



ddB0515 on Jan 6



That's excellent news man 💪 well done
I hope for frame-buffer in 2022 :D



Write a reply

Write

Preview

Aa ^



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